MTH209: Worksheet 1

Matrix Computations

1. Define any 3×3 matrices A and B of your choice and verify that

$$AB = \sum_{i=1}^{3} a_{.i} b_{i.} \,,$$

where $a_{.i}$ is the ith column of A and $b_{i.}$ is the ith row of B.

2. Study the code below:

```
n <- 100
A <- matrix(runif(n^2), nrow = n, ncol = n)
time <- system.time(A%*%A)[3]
```

What is the code doing?

3. Run Q2 for different values of `n = c(1e2, 5e2, 1e3, 2e3, 3e3, 4e3, 5e3)` and plot time versus n. Can you explain the behavior?

4. Matrix inversion can be done through a few different methods:

```
n <- 100
A <- matrix(runif(n^2), nrow = n, ncol = n)
A <- t(A)%*% A

## standard
A.inv <- solve(A)

## qr solve
A.inv <- qr.solve(A)

## cholesky
# Cholesky decomposition
R <- chol(A)        # A = R^T R
# Inverse via Cholesky
A.inv <- chol2inv(R)

## Singular Value Decomposition
svd_A <- svd(A)
```

```r
U <- svd_A$u
D <- svd_A$d
V <- svd_A$v

# Inverse via SVD
A.inv <- V %*% diag(1 / D) %*% t(U)
```

Determine which method is fastest?

5. Implement all matrix inversion methods for the following matrix

```r
# Create an ill-conditioned SPD matrix
n <- 8
A <- matrix(0, n, n)
for (i in 1:n) {
  for (j in 1:n) {
    A[i, j] <- 1 / (i + j - 1)
  }
}

# Check symmetry
# isSymmetric(A)
```

Determine which method gives an accurate inverse and which does not?